

Notes: Abel and Eberly (AER, 1994)

A Unified Model of Investment Under Uncertainty

Contents

1	Big picture	3
2	Incremental contribution of the paper	3
3	Model ingredients	4
3.1	Operating profit and uncertainty	4
3.2	Capital accumulation	4
3.3	Three cost components	5
3.4	Augmented adjustment cost and the investment dummy	5
4	Methodology and core theoretical result	6
4.1	Firm value and the Bellman equation	6
4.2	Investment conditional on paying the fixed cost	6
4.3	Choosing whether to be active	6
4.4	Three investment regimes	7
5	Interpretation of the main result	7
5.1	Why investment is nondecreasing in q	7
5.2	What generates the inaction interval	7
5.3	Irreversibility as observational equivalence	8
5.4	Discontinuities in investment	8
5.5	Relation to standard q -theory	8
6	What is q?	9
6.1	Euler-equation interpretation	9
6.2	Present-value interpretation	9
6.3	Why q is hard to use empirically	9
7	Relating q to observable variables	9
7.1	Setup for the examples	9
7.2	Case I: linearly homogeneous augmented adjustment costs	10
7.3	What the homogeneity assumption means	10

7.4	Case II: adjustment costs independent of capital	10
7.5	Cobb-Douglas example with stochastic profitability	11
7.6	Uncertainty and investment	11
8	Competitive-equilibrium discussion	11
8.1	Pindyck's critique	11
8.2	Response for Case I	12
8.3	Response for Case II	12
9	Identification, interpretation, and limits	12
9.1	What the paper establishes	12
9.2	Why marginal q , not average q , is fundamental	12
9.3	What the thresholds depend on	12
9.4	Main limitation	13
9.5	Future-work direction in the paper	13
10	Figures	13
10.1	Figure 1: The reward to investing	13
11	Short summary	14
12	Conclusion	15
12.1	Core conclusion	15
12.2	Economic intuition	15
12.3	Incremental contribution revisited	15
12.4	Implications	15
12.5	Limitations	15
12.6	Takeaway	15

1 Big picture

- **Research question.** How should a firm choose investment when adjustment is costly, capital may be partially irreversible, and investment may involve fixed costs?
- **Core idea.** The paper builds a single investment model that nests standard convex adjustment-cost models, irreversible-investment models, and models with fixed investment costs.
- **Main object.** The central state variable for investment is q , the shadow price of installed capital, also called marginal q .
- **Main result.** Investment is a nondecreasing function of q , but the policy is not necessarily a smooth upward-sloping curve. Instead, there can be three regimes: negative gross investment, zero investment, and positive gross investment.
- **Two critical thresholds.** The policy is summarized by two critical values, $q_1 < q_2$. If $q < q_1$, the firm sells or sheds capital; if $q_1 \leq q \leq q_2$, the firm does nothing; if $q > q_2$, the firm invests.
- **Why inaction arises.** A nondegenerate zero-investment region arises from fixed costs or from a kink in the augmented adjustment-cost function at zero investment.
- **Irreversibility as an outcome.** Irreversibility is not simply assumed. It can emerge endogenously when selling capital is sufficiently costly, so that negative investment is never optimal.
- **Economic takeaway.** The familiar q -theory insight survives: investment rises with marginal q . But with realistic frictions, the response has thresholds, inaction, and sometimes an effective irreversibility constraint.

2 Incremental contribution of the paper

- **Unification of two literatures.** Earlier adjustment-cost models emphasized smooth convex costs and positive investment, while irreversibility models emphasized waiting and nonnegative investment. Abel and Eberly put these cases in one framework.
- **Fixed costs are made formal.** The paper explicitly incorporates fixed costs of investment, which had been mentioned in earlier work but rarely analyzed formally.
- **Purchase price versus resale price.** The model allows the purchase price of capital to exceed the resale price. This wedge captures firm specificity, resale discounts, adverse selection in used capital markets, and other sources of partial irreversibility.
- **Irreversibility without infinite costs.** The paper shows that irreversible-looking behavior does not require infinite costs of negative investment. It is enough that the augmented adjustment cost does not make selling installed capital privately attractive.

- **A transparent threshold characterization.** Instead of deriving a separate policy for every friction, the paper characterizes investment through the roots of a reward function $\psi(q, K)$.
- **Connection to observable q .** Because marginal q is generally unobservable, the paper gives special cases in which marginal q can be related to observable average q or to observable product-market variables.
- **Clarification of uncertainty effects.** The paper shows that uncertainty can raise investment through its effect on q , but only weakly when the firm is inside the inaction region.
- **Competitive-equilibrium caveat.** The paper discusses why industry equilibrium and endogenous prices matter for interpreting uncertainty effects in competitive industries.

3 Model ingredients

3.1 Operating profit and uncertainty

The firm uses installed capital K_t and costlessly adjustable inputs, such as labor, to produce output. At each date, it chooses the flexible inputs to maximize instantaneous operating profit. Let

$$\pi(K_t, \varepsilon_t)$$

denote this maximized operating profit, where ε_t is an exogenous stochastic state variable that can represent technology, input prices, or demand conditions.

The stochastic state evolves as a diffusion:

$$d\varepsilon_t = \mu(\varepsilon_t)dt + \sigma(\varepsilon_t)dz_t,$$

where z_t is a standard Wiener process.

- $\pi_K(K, \varepsilon) > 0$: more installed capital raises operating profit.
- $\pi_{KK}(K, \varepsilon) < 0$: the marginal operating profit of capital is diminishing.
- The uncertainty is general at first; the paper later gives more specific examples to relate q to observable variables.

3.2 Capital accumulation

Gross investment is I_t , and capital depreciates at rate δ . Capital evolves according to

$$dK_t = (I_t - \delta K_t)dt.$$

- Positive I_t means purchasing and installing new capital.
- Zero I_t means allowing the capital stock to decline only through depreciation.

- Negative I_t means selling, scrapping, or otherwise reducing installed capital faster than depreciation.

3.3 Three cost components

The paper's key modeling move is to combine three kinds of investment frictions into an **augmented adjustment-cost function**.

1. **Purchase and sale costs.** The firm can buy uninstalled capital at price p_K^+ and sell it at price p_K^- , with

$$p_K^+ \geq p_K^- \geq 0.$$

A wedge between p_K^+ and p_K^- captures partial irreversibility.

2. **Convex adjustment costs.** These are installation, organizational, or disruption costs that are minimized at zero investment and become more severe as investment moves away from zero.
3. **Fixed costs.** These are costs incurred whenever investment is nonzero. The firm avoids them by setting $I = 0$.

3.4 Augmented adjustment cost and the investment dummy

Let $c(I, K)$ denote the augmented adjustment cost. The total cost paid by the firm is

$$\nu c(I, K),$$

where $\nu = 0$ if $I = 0$ and $\nu = 1$ if $I \neq 0$.

This dummy variable matters because fixed costs are not paid when the firm does nothing. Thus $c(0, K)$ is the limiting cost of making an arbitrarily small investment, not the cost actually paid at $I = 0$.

- $c(I, K)$ is continuous and strictly convex in I , except possibly at $I = 0$.
- The function may have a kink at $I = 0$.
- The one-sided derivatives at zero, $c_I(0, K)^-$ and $c_I(0, K)^+$, need not be equal.
- The fixed cost creates a discontinuity in the total investment-cost schedule even if $c(I, K)$ itself is continuous.

4 Methodology and core theoretical result

4.1 Firm value and the Bellman equation

The firm is risk neutral and maximizes the expected present value of operating profits net of total investment costs:

$$V(K_t, \varepsilon_t) = \max_{\{I, \nu\}} E_t \int_0^\infty [\pi(K_{t+s}, \varepsilon_{t+s}) - \nu_{t+s} c(I_{t+s}, K_{t+s})] e^{-rs} ds.$$

Let

$$q = V_K(K, \varepsilon)$$

be the shadow value of one additional installed unit of capital. The Bellman equation can be written as

$$rV = \max_{I, \nu} \left\{ \pi(K, \varepsilon) - \nu c(I, K) + q(I - \delta K) + \mu(\varepsilon)V_\varepsilon + \frac{1}{2}\sigma^2(\varepsilon)V_{\varepsilon\varepsilon} \right\}.$$

The only terms involving I and ν are $-\nu c(I, K) + qI$. Therefore, the investment problem reduces to

$$\max_{I, \nu} [qI - \nu c(I, K)].$$

4.2 Investment conditional on paying the fixed cost

First suppose $\nu = 1$. Define the reward from active investment as

$$\psi(q, K) = \max_I [qI - c(I, K)].$$

Let $I^*(q, K)$ be the investment level that solves this problem when the firm is active.

When the optimum is away from the kink at zero, the first-order condition is

$$c_I(I^*(q, K), K) = q.$$

Because $c(I, K)$ is strictly convex in I , $I^*(q, K)$ is increasing in q . If the marginal cost schedule has a gap at zero, then the conditional optimum is $I^*(q, K) = 0$ for values of q inside that gap.

4.3 Choosing whether to be active

If the firm sets $\nu = 0$, it sets $I = 0$, pays no fixed cost, and obtains value zero from the investment subproblem. If it sets $\nu = 1$, it obtains $\psi(q, K)$.

Therefore:

- active positive or negative investment occurs only when $\psi(q, K) > 0$;
- inaction occurs when $\psi(q, K) \leq 0$.

Let q_1 and q_2 be the smallest and largest roots of

$$\psi(q, K) = 0.$$

These thresholds summarize the entire investment policy.

4.4 Three investment regimes

The optimal gross investment rule is

$$I(q, K) = \begin{cases} I^*(q, K) < 0, & q < q_1, \\ 0, & q_1 \leq q \leq q_2, \\ I^*(q, K) > 0, & q > q_2. \end{cases}$$

- **Negative investment region:** $q < q_1$. Installed capital is sufficiently unattractive that the firm sells, scraps, or reduces capital.
- **Inaction region:** $q_1 \leq q \leq q_2$. The value of adjusting does not cover the fixed, kink, or resale-friction costs.
- **Positive investment region:** $q > q_2$. Installed capital is sufficiently valuable that the firm expands.

Economic message. The model creates an S, s -style investment policy in the language of marginal q . The firm waits over an interval of shadow values and adjusts only when the value of adjustment is large enough.

5 Interpretation of the main result

5.1 Why investment is nondecreasing in q

The variable q is the marginal value of installed capital. A higher q raises the marginal benefit of investment qI , making positive investment more attractive and negative investment less attractive. Strict convexity of $c(I, K)$ ensures monotonicity.

- In the positive-investment region, higher q means a higher optimal I .
- In the inaction region, higher q may not change investment at all.
- In the negative-investment region, higher q reduces the incentive to sell capital.

5.2 What generates the inaction interval

There is a nondegenerate inaction region when at least one of two frictions is present:

- **Fixed investment costs.** If $c(0, K) > 0$, the firm must pay a fixed cost whenever $I \neq 0$, so small adjustments are not worthwhile.
- **A kink at zero.** If $c_I(0, K)^- < c_I(0, K)^+$, there is a range of marginal valuations for which zero investment is optimal.

If there are no fixed costs and the cost function is differentiable at zero, then the inaction range collapses to a single point, reproducing the smooth adjustment-cost model.

5.3 Irreversibility as observational equivalence

The paper's treatment of irreversibility is especially important. It does not simply impose $I \geq 0$. Instead, it asks when the firm's optimal policy will never choose negative investment. If

$$\min_I c(I, K) \geq 0,$$

then negative investment is never optimal. Intuitively, the firm gives up installed capital that has positive value. To make selling worthwhile, the augmented adjustment cost would have to be negative, meaning that net resale proceeds exceed the relevant sale and adjustment costs. If that never happens, the firm behaves as if investment is irreversible.

Economic message. A model can look irreversible to an outside observer even though negative investment is technically allowed. The key is whether selling installed capital creates enough net private value.

5.4 Discontinuities in investment

When fixed costs are positive, the investment policy can jump at the thresholds:

- at q_1 , the firm jumps from negative investment to zero investment;
- at q_2 , the firm jumps from zero investment to positive investment.

This is why the model can generate lumpy investment. The firm waits, then undertakes a discrete adjustment once q crosses a threshold.

5.5 Relation to standard q -theory

Standard convex adjustment-cost models typically imply a smooth relationship between investment and marginal q . Abel and Eberly keep the central role of marginal q , but show that realistic costs alter the shape of the policy function.

- **Standard model:** investment responds smoothly to q .
- **This model:** investment is weakly increasing in q , but may feature inaction and jumps.
- **Irreversibility model:** the negative-investment region disappears when sale is unattractive enough.

6 What is q ?

6.1 Euler-equation interpretation

The paper derives an Euler equation for marginal q :

$$(r + \delta)q = \pi_K(K, \varepsilon) - \nu c_K(I, K) + \frac{E\{dq\}}{dt}.$$

The left-hand side is the required return on the marginal installed unit of capital. The right-hand side is the expected return from holding that unit:

- the marginal operating profit $\pi_K(K, \varepsilon)$;
- the marginal effect on adjustment costs, $-\nu c_K(I, K)$;
- the expected capital gain in q , $E\{dq\}/dt$.

6.2 Present-value interpretation

Using a Feynman-Kac argument, the paper shows that marginal q is the expected present value of future marginal profits of installed capital:

$$q_t = E_t \int_0^\infty [\pi_K(K_{t+s}, \varepsilon_{t+s}) - u_{t+s} c_K(I_{t+s}, K_{t+s})] e^{-(r+\delta)s} ds.$$

Economic message. Marginal q is forward-looking. It summarizes all future expected operating profits and future adjustment-cost effects associated with an extra unit of installed capital.

6.3 Why q is hard to use empirically

Marginal q is conceptually central but generally unobservable. Observed market value gives average q , not marginal q . Therefore, the paper gives special cases in which marginal q can be linked to variables that researchers may observe.

7 Relating q to observable variables

7.1 Setup for the examples

The paper restricts attention to a competitive firm with a linearly homogeneous production function in capital and flexible inputs. After optimizing over flexible inputs, operating profit can be written as

$$\pi(K, \varepsilon) = H(\varepsilon)K.$$

This form means that operating profit is linear in the capital stock. The examples then differ in the assumptions placed on the augmented adjustment-cost function.

7.2 Case I: linearly homogeneous augmented adjustment costs

Suppose that both operating profit and the augmented adjustment-cost function are homogeneous of degree one in I and K . Then firm value is proportional to capital:

$$V(K, \varepsilon) = q(\varepsilon)K.$$

Therefore,

$$q(\varepsilon) = \frac{V(K, \varepsilon)}{K}.$$

- Marginal q equals average q .
- Average q can be approximated using security-market values and replacement cost.
- This extends Hayashi's average- q result to a stochastic setting with fixed costs and irreversibility.

Economic message. Under strong homogeneity assumptions, the empirical object used in much of the q -theory literature can still be the correct shadow price guiding investment.

7.3 What the homogeneity assumption means

Linearly homogeneous augmented costs are plausible when each cost component scales with the size of the firm:

- purchase and sale costs scale with the quantity of capital transacted;
- standard installation costs are often modeled as functions of I/K multiplied by K ;
- fixed costs can be proportional to the scale of operations, for example if they represent the opportunity cost of shutting down production while investment occurs.

In this case, the optimal investment-capital ratio depends only on q :

$$\frac{I}{K} = G'^{-1}(q)$$

outside the inaction interval, with $I/K = 0$ inside it.

7.4 Case II: adjustment costs independent of capital

Now suppose that

$$c_K(I, K) = 0.$$

The augmented adjustment cost does not depend on the capital stock. The value function then takes the affine form

$$V(K, \varepsilon) = q(\varepsilon)K + J(\varepsilon),$$

where $J(\varepsilon) > 0$.

Here average q is not generally equal to marginal q , because $V/K = q + J/K$. Instead, the paper derives q from product-market primitives in a more specialized example.

7.5 Cobb-Douglas example with stochastic profitability

Let the competitive firm produce with a Cobb-Douglas technology using labor and capital. After optimizing over labor, operating profit becomes

$$\pi(K, p) = hp^\theta K,$$

where p is the product of output price and productivity, $h > 0$, and $\theta > 1$.

Assume p follows geometric Brownian motion:

$$\frac{dp}{p} = \sigma dz.$$

Then marginal q is

$$q_t = \frac{hp_t^\theta}{r + \delta - \frac{1}{2}\theta(\theta - 1)\sigma^2},$$

provided the denominator is positive.

7.6 Uncertainty and investment

Because $\theta > 1$, an increase in σ increases q_t for a given p_t . This reproduces the classic result that uncertainty can raise investment by increasing the value of the marginal product stream.

But the paper's richer policy function adds an important qualification:

- if $q_t < q_1$ or $q_t > q_2$, the increase in q_t changes investment;
- if $q_t \in [q_1, q_2]$, a small increase in q_t may leave the firm in the inaction region, so investment remains zero.

Economic message. Uncertainty may raise the shadow value of capital, but actual investment can remain unchanged when fixed costs or irreversibility put the firm in a waiting region.

8 Competitive-equilibrium discussion

8.1 Pindyck's critique

The paper discusses two issues raised by Pindyck's competitive-equilibrium critique.

- **Adjustment costs and firm size.** If adjustment costs depend only on I , a competitive industry with free entry might fragment into many tiny firms, making adjustment costs irrelevant in the limit.
- **Endogenous output prices.** Even competitive firms take prices as given, but the industry does not. An industry-wide shock can change aggregate output and therefore equilibrium prices.

8.2 Response for Case I

In Case I, the augmented adjustment-cost function is homogeneous in I and K . Firm value is exactly proportional to capital, so a zero-capital entrant has zero value. There is no positive entry value $J(\varepsilon)$ to exploit by dividing the firm into arbitrarily small pieces.

Economic message. Pindyck's irrelevance argument does not apply to the homogeneous augmented-cost case because marginal adjustment costs remain relevant even as firm scale changes.

8.3 Response for Case II

In Case II, the uncertainty result depends on what kind of shock drives p .

- If p reflects an industry-wide output-price or demand shock, equilibrium price responses may offset or reverse the partial-equilibrium effect.
- If p reflects an idiosyncratic productivity shock to one competitive firm, the firm takes the output price as given and the paper's partial-equilibrium result is more appropriate.

Economic message. The model is a firm-level investment model. Applying it to competitive industries requires care about whether shocks are idiosyncratic or aggregate.

9 Identification, interpretation, and limits

9.1 What the paper establishes

The paper establishes a theoretical characterization of investment under uncertainty with a broad class of frictions. It does not estimate the model, but it provides the policy-function logic that later empirical work can use.

9.2 Why marginal q , not average q , is fundamental

Investment depends on the value of an additional installed unit of capital. That is marginal q . Average q is useful only under restrictions that make it equal or proportional to marginal q .

9.3 What the thresholds depend on

The thresholds q_1 and q_2 depend only on the augmented adjustment-cost function. They do not depend directly on the operating-profit function or on the stochastic process for ε . Those primitives determine where q is likely to be, while the cost function determines how the firm reacts once q is known.

9.4 Main limitation

The most important practical limitation is that the core state variable, marginal q , is generally unobservable. The paper provides special cases to connect q to observed variables, but the general model still needs an empirical mapping from data to marginal q .

9.5 Future-work direction in the paper

The authors note that ongoing work studies mean-reverting profitability processes so that q has a stationary distribution. The goal is to derive econometric specifications that can be taken to aggregate and disaggregate investment data.

10 Figures

10.1 Figure 1: The reward to investing

Read:

- The vertical axis is the reward $\psi(q, K)$ from choosing active investment rather than doing nothing.
- The horizontal axis is marginal q , the shadow value of installed capital.
- The curve is convex and reaches a flat bottom at $-c(0, K)$ when q lies between the left and right marginal costs around zero investment.
- The roots q_1 and q_2 determine the thresholds for negative investment and positive investment.
- Between q_1 and q_2 , the reward from active adjustment is not positive, so the firm chooses zero gross investment.

Economic message: The figure is the central visualization of the paper. It shows how fixed costs and kinks convert a smooth q -investment relation into a threshold policy with a region of inaction.

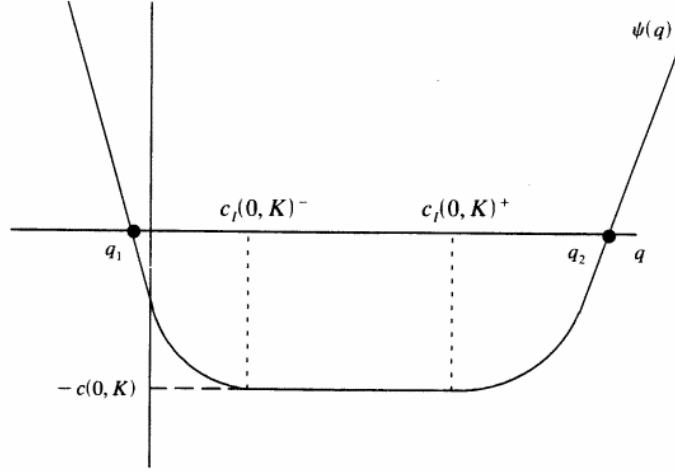


FIGURE 1. THE REWARD TO INVESTING

Figure 1: The reward to investing.

11 Short summary

- Abel and Eberly build a unified theory of investment under uncertainty with convex adjustment costs, fixed costs, purchase-sale wedges, and irreversibility.
- The key state variable is marginal q , the shadow price of installed capital.
- The optimal investment rule is monotone in q , but not necessarily smooth.
- Two thresholds, q_1 and q_2 , divide the policy into negative investment, zero investment, and positive investment.
- Fixed costs and kinks generate a nondegenerate inaction region.
- Irreversibility can emerge endogenously when selling installed capital never creates enough net value to justify negative investment.
- Marginal q is the expected present value of future marginal operating profits net of adjustment-cost effects.
- Under homogeneity, marginal q equals average q , making Tobin's q observable and useful.
- In a Cobb-Douglas example with stochastic profitability, higher uncertainty raises q , but investment may remain unchanged inside the inaction region.
- The competitive-equilibrium discussion emphasizes that industry-wide shocks require care because equilibrium output prices may adjust.

12 Conclusion

12.1 Core conclusion

The paper's core conclusion is that the q -theory of investment survives in a much richer environment, but it must be interpreted as a threshold theory rather than a purely smooth adjustment theory. Investment is increasing in marginal q , yet fixed costs, resale wedges, and irreversibility create inaction and discontinuous adjustment.

12.2 Economic intuition

A firm invests when the shadow value of installed capital is high enough to justify the purchase and adjustment costs. It sells capital only when the shadow value is low enough and resale proceeds are attractive enough. When the shadow value is in the middle, the firm waits because small adjustments do not cover fixed or kink costs.

12.3 Incremental contribution revisited

The paper brings together three investment frictions that were often treated separately: convex adjustment costs, fixed costs, and irreversibility. It shows that all of them can be represented through the augmented adjustment-cost function and summarized by two q -thresholds.

12.4 Implications

The model helps explain why investment is often lumpy and inactive for long periods even when fundamentals move continuously. It also explains why empirical investment equations based only on smooth q -responses may miss important nonlinearities.

12.5 Limitations

The model is theoretical and relies on marginal q , which is generally hard to observe. The examples connecting q to data require strong assumptions, such as homogeneity or specific stochastic processes. Competitive-equilibrium effects also matter when shocks are common to an industry.

12.6 Takeaway

The enduring takeaway is that investment under uncertainty is governed by marginal q , but realistic frictions transform the investment rule into a band of inaction with thresholds for expansion and contraction. This unified view links smooth adjustment-cost models, irreversible-investment models, and fixed-cost models in one framework.